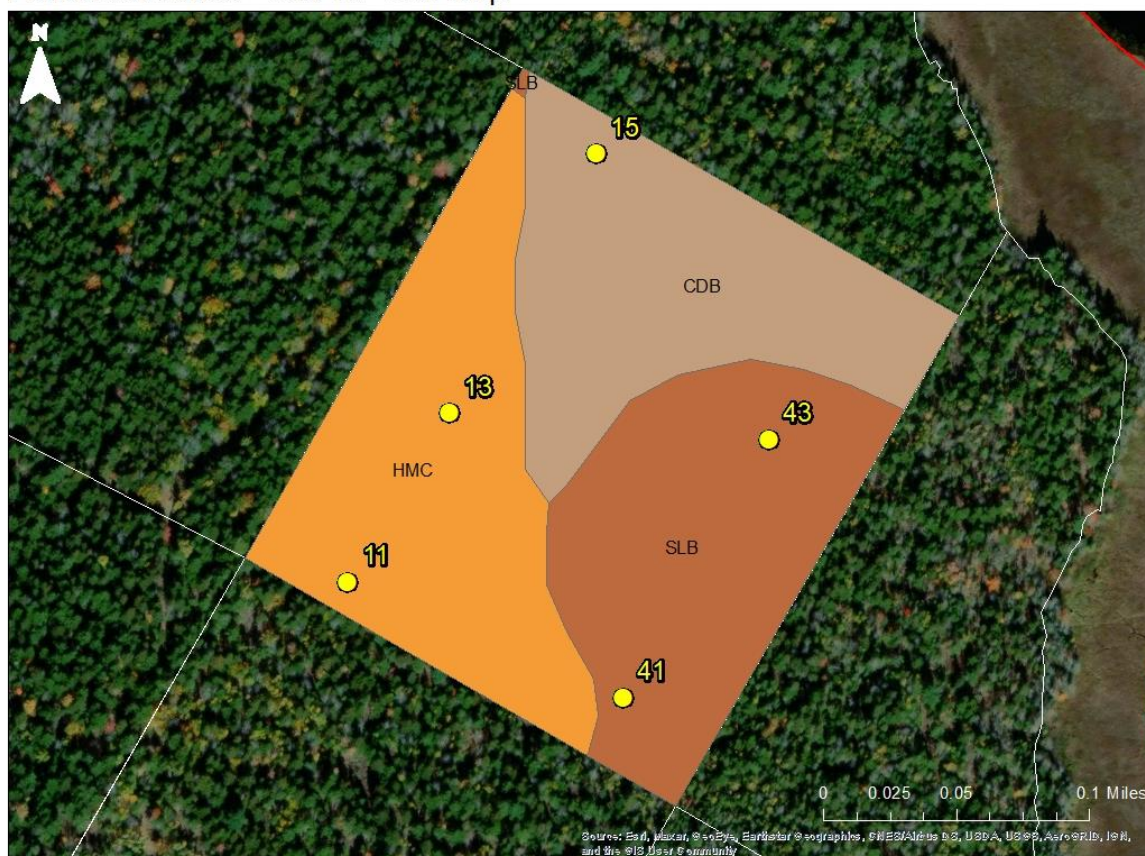


## Penobscot Soil and Environmental Report

| Penobscot County, Maine, Southern Part (ME612) |                                                                        |              |                |
|------------------------------------------------|------------------------------------------------------------------------|--------------|----------------|
| Map Unit Symbol                                | Map Unit Name                                                          | Acres in AOI | Percent of AOI |
| CDB                                            | Colonel-Peru-Brayton complex, 0 to 8 percent slopes, very stony        | 8.1          | 31.7%          |
| HMC                                            | Monadnock-Hermon complex, 3 to 15 percent slopes, extremely bouldery   | 9.1          | 35.8%          |
| SLB                                            | Scantic-Lamoine-Colonel association, 0 to 8 percent slopes, very stony | 8.3          | 32.5%          |
| <b>Totals for Area of Interest</b>             |                                                                        | <b>25.5</b>  | <b>100.0%</b>  |

Penobscot Forest - Unit 15 - Soil Map



**CDB—Colonel-Peru-Brayton complex, 0 to 8 percent slopes, very stony**

**Map Unit Setting**

- *National map unit symbol:* szrs
- *Elevation:* 50 to 1,000 feet
- *Mean annual precipitation:* 39 to 45 inches
- *Mean annual air temperature:* 42 to 45 degrees F
- *Frost-free period:* 115 to 155 days
- *Farmland classification:* Not prime farmland

**Map Unit Composition**

- *Colonel and similar soils:* 45 percent
- *Peru and similar soils:* 30 percent
- *Brayton and similar soils:* 15 percent
- *Minor components:* 10 percent

**Description of Colonel**

**Setting**

- *Landform:* Ground moraines
- *Landform position (two-dimensional):* Footslope, toeslope
- *Landform position (three-dimensional):* Base slope
- *Down-slope shape:* Linear
- *Across-slope shape:* Linear
- *Parent material:* Coarse-loamy lodgment till

**Typical profile**

- *Oa - 0 to 3 inches:* highly decomposed plant material
- *E - 3 to 5 inches:* fine sandy loam
- *Bs - 5 to 19 inches:* fine sandy loam
- *BC - 19 to 22 inches:* sandy loam
- *Cd - 22 to 65 inches:* sandy loam

**Properties and qualities**

- *Slope:* 2 to 6 percent
- *Surface area covered with cobbles, stones or boulders:* 1.5 percent
- *Depth to restrictive feature:* 14 to 23 inches to densic material
- *Drainage class:* Somewhat poorly drained
- *Capacity of the most limiting layer to transmit water (Ksat):* Moderately low to moderately high (0.01 to 1.42 in/hr)
- *Depth to water table:* About 6 to 20 inches
- *Frequency of flooding:* None
- *Frequency of ponding:* None
- *Maximum salinity:* Nonsaline (0.0 to 1.9 mmhos/cm)
- *Available water capacity:* Low (about 4.1 inches)

**Description of Peru**

**Setting**

- *Landform:* Ground moraines
- *Landform position (two-dimensional):* Footslope, backslope
- *Landform position (three-dimensional):* Base slope, side slope
- *Down-slope shape:* Linear
- *Across-slope shape:* Linear
- *Parent material:* Coarse-loamy lodgment till

**Typical profile**

- *Oa - 0 to 2 inches:* highly decomposed plant material
- *E - 2 to 3 inches:* fine sandy loam
- *Bs - 3 to 11 inches:* fine sandy loam
- *BC - 11 to 26 inches:* fine sandy loam
- *Cd - 26 to 65 inches:* gravelly fine sandy loam

**Properties and qualities**

- *Slope:* 5 to 8 percent
- *Surface area covered with cobbles, stones or boulders:* 1.5 percent
- *Depth to restrictive feature:* 21 to 33 inches to densic material
- *Drainage class:* Moderately well drained
- *Capacity of the most limiting layer to transmit water (Ksat):* Very low to moderately high (0.00 to 1.42 in/hr)
- *Depth to water table:* About 16 to 28 inches
- *Frequency of flooding:* None
- *Frequency of ponding:* None
- *Maximum salinity:* Nonsaline (0.0 to 1.9 mmhos/cm)
- *Available water capacity:* Low (about 4.5 inches)

## Description of Brayton

### Setting

- *Landform*: Ground moraines
- *Landform position (two-dimensional)*: Toeslope, footslope
- *Landform position (three-dimensional)*: Base slope
- *Down-slope shape*: Concave
- *Across-slope shape*: Linear
- *Parent material*: Loamy lodgment till

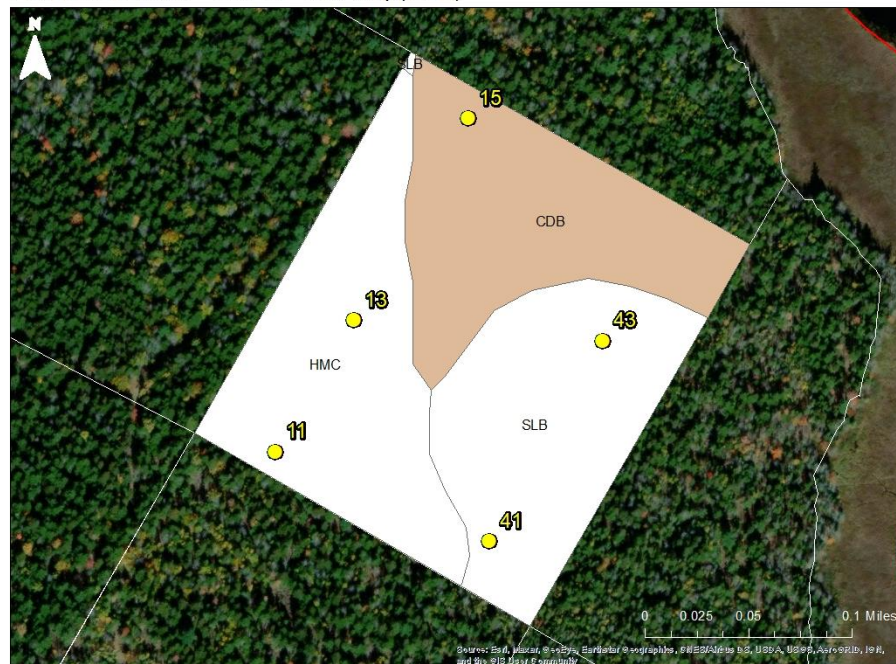
### Typical profile

- *Oa - 0 to 5 inches*: muck
- *Eg - 5 to 10 inches*: gravelly fine sandy loam
- *Bg - 10 to 16 inches*: fine sandy loam
- *BC - 16 to 23 inches*: fine sandy loam
- *Cd - 23 to 65 inches*: fine sandy loam

### Properties and qualities

- *Slope*: 0 to 4 percent
- *Surface area covered with cobbles, stones or boulders*: 1.5 percent
- *Depth to restrictive feature*: 16 to 25 inches to densic material
- *Drainage class*: Poorly drained
- *Capacity of the most limiting layer to transmit water (Ksat)*: Moderately low to moderately high (0.01 to 1.42 in/hr)
- *Depth to water table*: About 0 to 12 inches
- *Frequency of flooding*: None
- *Frequency of ponding*: None
- *Maximum salinity*: Nonsaline (0.0 to 1.9 mmhos/cm)
- *Available water capacity*: Low (about 4.2 inches)

Penobscot Forest - Unit 15 - Soil Map(CDB)



## **HMC—Monadnock-Hermon complex, 3 to 15 percent slopes, extremely bouldery**

- *National map unit symbol:* 2x9pv
- *Elevation:* 30 to 490 feet
- *Mean annual precipitation:* 31 to 65 inches
- *Mean annual air temperature:* 36 to 52 degrees F
- *Frost-free period:* 90 to 160 days
- *Farmland classification:* Not prime farmland
- *Monadnock, extremely bouldery, and similar soils:* 45 percent
- *Hermon, extremely bouldery, and similar soils:* 40 percent
- *Minor components:* 15 percent

### **Description of **Monadnock**, Extremely Bouldery**

#### **Setting**

- *Landform:* Hills
- *Landform position (two-dimensional):* Summit, shoulder backslope
- *Landform position (three-dimensional):* Side slope, nose slope, interflue
- *Down-slope shape:* Convex
- *Across-slope shape:* Convex
- *Parent material:* Loamy supraglacial meltout till derived from granite and gneiss and/or mica schist and/or phyllite over sandy and gravelly supraglacial meltout till derived from granite and gneiss and/or mica schist and/or phyllite

#### **Typical profile**

- *Oe - 0 to 3 inches:* moderately decomposed plant material
- *E - 3 to 8 inches:* fine sandy loam
- *Bs1 - 8 to 10 inches:* fine sandy loam
- *Bs2 - 10 to 12 inches:* fine sandy loam
- *Bs3 - 12 to 22 inches:* gravelly fine sandy loam
- *BC - 22 to 25 inches:* gravelly fine sandy loam
- *2C1 - 25 to 45 inches:* gravelly loamy sand
- *2C2 - 45 to 65 inches:* gravelly loamy sand

#### **Properties and qualities**

- *Slope:* 3 to 15 percent
- *Surface area covered with cobbles, stones or boulders:* 6.0 percent
- *Depth to restrictive feature:* 18 to 36 inches to strongly contrasting textural stratification
- *Drainage class:* Well drained
- *Capacity of the most limiting layer to transmit water (Ksat):* Moderately low to high (0.14 to 14.03 in/hr)
- *Depth to water table:* More than 80 inches
- *Frequency of flooding:* None
- *Frequency of ponding:* None
- *Maximum salinity:* Nonsaline (0.0 to 1.9 mmhos/cm)
- *Available water capacity:* Low (about 4.3 inches)

### **Description of **Hermon**, Extremely Bouldery**

#### **Setting**

- *Landform:* Hills
- *Landform position (two-dimensional):* Backslope, summit, shoulder
- *Landform position (three-dimensional):* Side slope, nose slope, interflue
- *Down-slope shape:* Convex
- *Across-slope shape:* Convex
- *Parent material:* Sandy and gravelly supraglacial meltout till derived from granite and gneiss

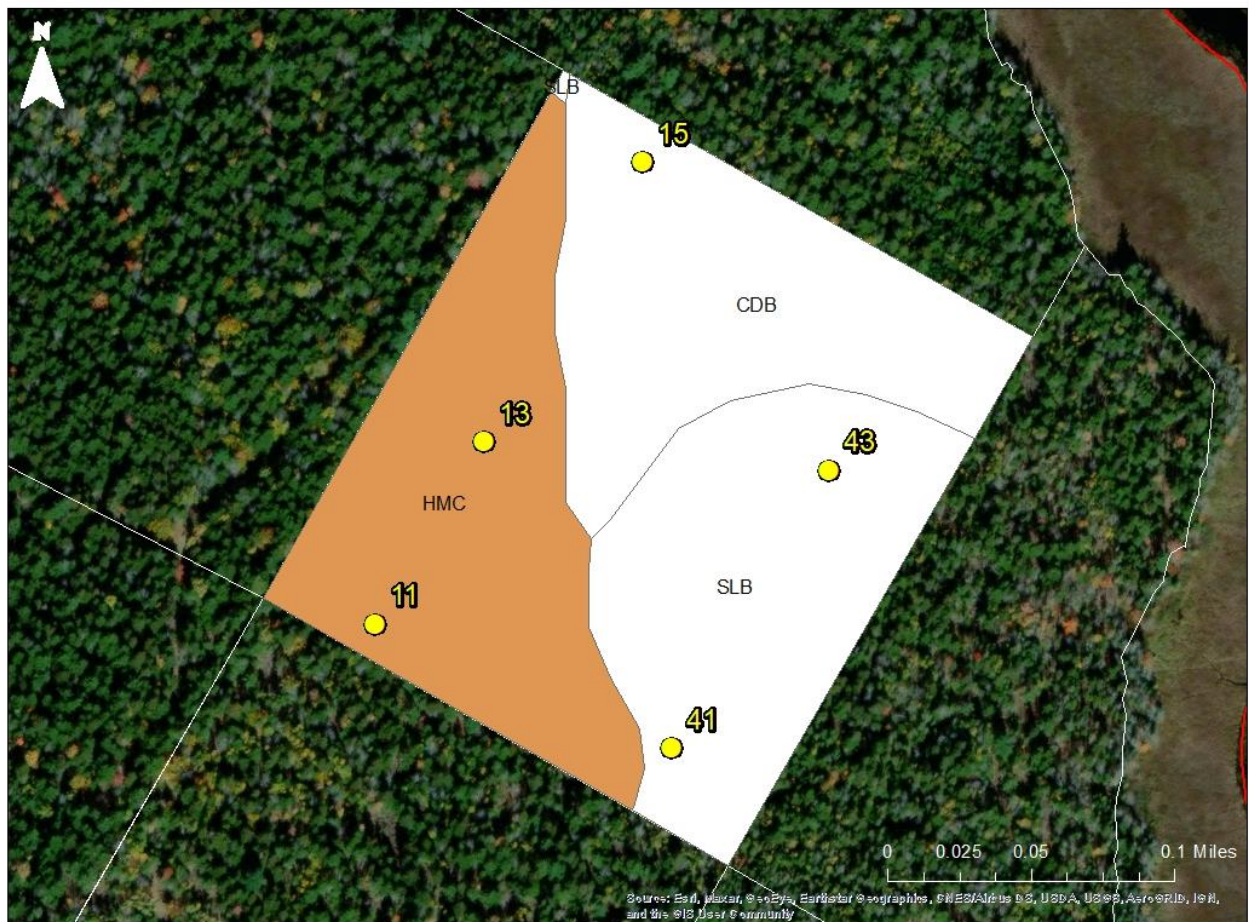
#### **Typical profile**

- *Oa - 0 to 2 inches:* highly decomposed plant material
- *E - 2 to 3 inches:* sandy loam
- *Bhs - 3 to 9 inches:* sandy loam
- *Bs1 - 9 to 16 inches:* very gravelly sandy loam
- *Bs2 - 16 to 32 inches:* extremely gravelly loamy sand
- *C - 32 to 65 inches:* very gravelly coarse sand

#### **Properties and qualities**

- *Slope:* 3 to 15 percent
- *Surface area covered with cobbles, stones or boulders:* 6.0 percent
- *Depth to restrictive feature:* More than 80 inches
- *Drainage class:* Somewhat excessively drained
- *Capacity of the most limiting layer to transmit water (Ksat):* Moderately high to high (1.42 to 14.03 in/hr)
- *Depth to water table:* More than 80 inches
- *Maximum salinity:* Nonsaline (0.0 to 1.9 mmhos/cm)
- *Available water capacity:* Low

# Penobscot Forest - Unit 15 - Soil Map(HMC)



**SLB—Scantic-Lamoine-Colonel association, 0 to 8 percent slopes, very stony**

*National map unit symbol: sx8z*

- *Elevation: 20 to 1,000 feet*
- *Mean annual precipitation: 39 to 45 inches*
- *Mean annual air temperature: 42 to 45 degrees F*
  - *Frost-free period: 115 to 155 days*
- *Farmland classification: Not prime farmland*
  - *Scantic and similar soils: 45 percent*
  - *Lamoine and similar soils: 25 percent*
  - *Colonel and similar soils: 15 percent*
  - *Minor components: 15 percent*

**Description of Scantic**

**Setting**

- *Landform: Marine terraces, river valleys*
- *Landform position (two-dimensional): Toeslope, footslope*
- *Landform position (three-dimensional): Base slope*
- *Down-slope shape: Linear*
- *Across-slope shape: Concave*
- *Parent material: Fine glaciomarine deposits*

**Typical profile**

- *Oa - 0 to 3 inches: mucky peat*
- *Eg - 3 to 9 inches: silty clay loam*
- *Bg - 9 to 30 inches: silty clay loam*
- *Cg - 30 to 65 inches: silty clay*

**Properties and qualities**

- *Slope: 0 to 4 percent*
- *Depth to restrictive feature: More than 80 inches*
- *Drainage class: Poorly drained*
- *Capacity of the most limiting layer to transmit water (Ksat): Very low to moderately low (0.00 to 0.14 in/hr)*
- *Depth to water table: About 0 to 12 inches*
- *Frequency of flooding: None*
- *Frequency of ponding: None*
- *Maximum salinity: Nonsaline (0.0 to 1.9 mmhos/cm)*
- *Available water capacity: Moderate (about 7.0 inches)*

**Description of Lamoine**

**Setting**

- *Landform: Marine terraces, river valleys*
- *Landform position (two-dimensional): Footslope, toeslope*
- *Landform position (three-dimensional): Base slope*
- *Down-slope shape: Linear*
- *Across-slope shape: Linear*
- *Parent material: Fine glaciomarine deposits*

**Typical profile**

- *Oe - 0 to 2 inches: moderately decomposed plant material*
- *A - 2 to 8 inches: silt loam*
- *Bw - 8 to 15 inches: silt loam*
- *Bg - 15 to 24 inches: silty clay loam*
- *Cg - 24 to 65 inches: silty clay*

**Properties and qualities**

- *Slope: 3 to 8 percent*
- *Depth to restrictive feature: More than 80 inches*
- *Drainage class: Somewhat poorly drained*
- *Capacity of the most limiting layer to transmit water (Ksat): Very low to moderately low (0.00 to 0.14 in/hr)*
- *Depth to water table: About 6 to 17 inches*
- *Frequency of flooding: None*
- *Frequency of ponding: None*
- *Maximum salinity: Nonsaline (0.0 to 1.9 mmhos/cm)*
- *Available water capacity: Moderate (about 7.7 inches)*

## Description of Colonel

### Setting

- *Landform*: Ground moraines
- *Landform position (two-dimensional)*: Footslope, toeslope
- *Landform position (three-dimensional)*: Base slope
- *Down-slope shape*: Linear
- *Across-slope shape*: Linear
- *Parent material*: Coarse-loamy lodgment till

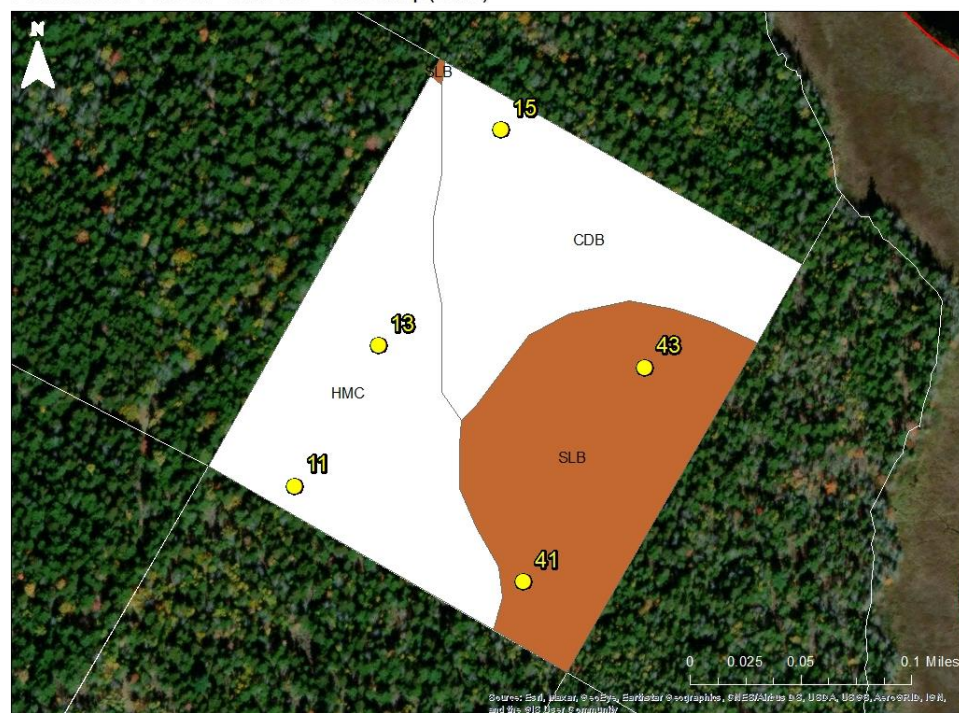
### Typical profile

- *Oa* - 0 to 3 inches: highly decomposed plant material
- *E* - 3 to 5 inches: fine sandy loam
- *Bs* - 5 to 19 inches: fine sandy loam
- *BC* - 19 to 22 inches: sandy loam
- *Cd* - 22 to 65 inches: sandy loam

### Properties and qualities

- *Slope*: 3 to 8 percent
- *Surface area covered with cobbles, stones or boulders*: 1.5 percent
- *Depth to restrictive feature*: 14 to 23 inches to densic material
- *Drainage class*: Somewhat poorly drained
- *Capacity of the most limiting layer to transmit water (Ksat)*: Moderately low to moderately high (0.01 to 1.42 in/hr)
- *Depth to water table*: About 6 to 20 inches
- *Frequency of flooding*: None
- *Frequency of ponding*: None
- *Maximum salinity*: Nonsaline (0.0 to 1.9 mmhos/cm)
- *Available water capacity*: Low (about 4.1 inches)

Penobscot Forest - Unit 15 - Soil Map(SLB)



**Tables — Forest Productivity (Tree Site Index): red maple (Carmean 1978 (095))**

| — Penobscot County, Maine, Southern Part (ME612) |                                                                        |               |              |                |
|--------------------------------------------------|------------------------------------------------------------------------|---------------|--------------|----------------|
| Map unit symbol                                  | Map unit name                                                          | Rating (feet) | Acres in AOI | Percent of AOI |
| CDB                                              | Colonel-Peru-Brayton complex, 0 to 8 percent slopes, very stony        | 65            | 8.1          | 31.7%          |
| HMC                                              | Monadnock-Hermon complex, 3 to 15 percent slopes, extremely bouldery   |               | 9.1          | 35.8%          |
| SLB                                              | Scantic-Lamoine-Colonel association, 0 to 8 percent slopes, very stony | 55            | 8.3          | 32.5%          |
| <b>Totals for Area of Interest</b>               |                                                                        |               | <b>25.5</b>  | <b>100.0%</b>  |

**Tables — pH (1 to 1 Water) — Summary By Map Unit**

| Summary by Map Unit — Penobscot County, Maine, Southern Part (ME612) |                                                                        |        |              |                |
|----------------------------------------------------------------------|------------------------------------------------------------------------|--------|--------------|----------------|
| Map unit symbol                                                      | Map unit name                                                          | Rating | Acres in AOI | Percent of AOI |
| CDB                                                                  | Colonel-Peru-Brayton complex, 0 to 8 percent slopes, very stony        | 5.2    | 8.1          | 31.7%          |
| HMC                                                                  | Monadnock-Hermon complex, 3 to 15 percent slopes, extremely bouldery   | 4.6    | 9.1          | 35.8%          |
| SLB                                                                  | Scantic-Lamoine-Colonel association, 0 to 8 percent slopes, very stony | 6.3    | 8.3          | 32.5%          |
| <b>Totals for Area of Interest</b>                                   |                                                                        |        | <b>25.5</b>  | <b>100.0%</b>  |

**Tables — Organic Matter — Summary By Map Unit**

| Summary by Map Unit — Penobscot County, Maine, Southern Part (ME612) |                                                                        |                  |              |                |
|----------------------------------------------------------------------|------------------------------------------------------------------------|------------------|--------------|----------------|
| Map unit symbol                                                      | Map unit name                                                          | Rating (percent) | Acres in AOI | Percent of AOI |
| CDB                                                                  | Colonel-Peru-Brayton complex, 0 to 8 percent slopes, very stony        | 3.98             | 8.1          | 31.7%          |
| HMC                                                                  | Monadnock-Hermon complex, 3 to 15 percent slopes, extremely bouldery   | 4.95             | 9.1          | 35.8%          |
| SLB                                                                  | Scantic-Lamoine-Colonel association, 0 to 8 percent slopes, very stony | 3.61             | 8.3          | 32.5%          |
| <b>Totals for Area of Interest</b>                                   |                                                                        |                  | <b>25.5</b>  | <b>100.0%</b>  |

